

Outset · AI in Healthcare · Day 3 · Group 2

Screening skin spots for melanoma

Can a model look at a photo of a skin spot and catch the melanomas before a dermatologist ever sees them?

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Outset

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BACKGROUND

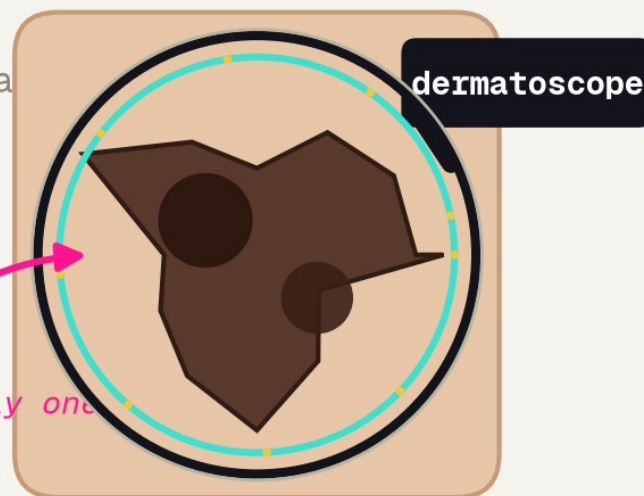
The spot that could be melanoma

Screening a skin spot: is it melanoma?

the 7 lesion types:

- actinic keratoses
- basal cell carcinoma
- benign keratosis
- dermatofibroma
- **MELANOMA**
- melanocytic nevi
- vascular lesions

the deadly one



lesion types from HAM10000 (Tschandl 2018)

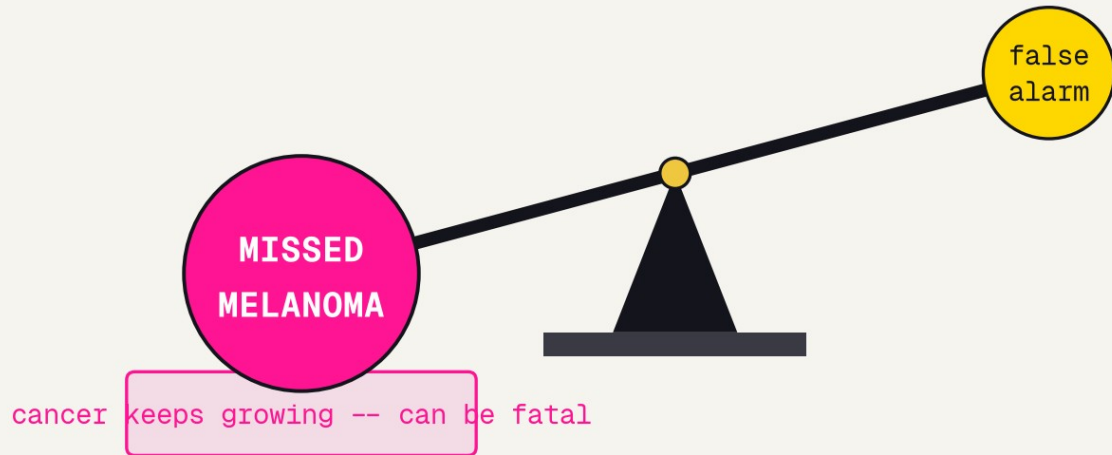
A dermatoscope magnifies the spot; the screen must sort seven look-alikes and never wave the melanoma through.

Why a miss is deadly

The two mistakes are not equal

so a screening tool leans toward catching cancers, even at the cost of false alarms

just one extra check-up



adapted from Esteva 2017; Haenssle 2018

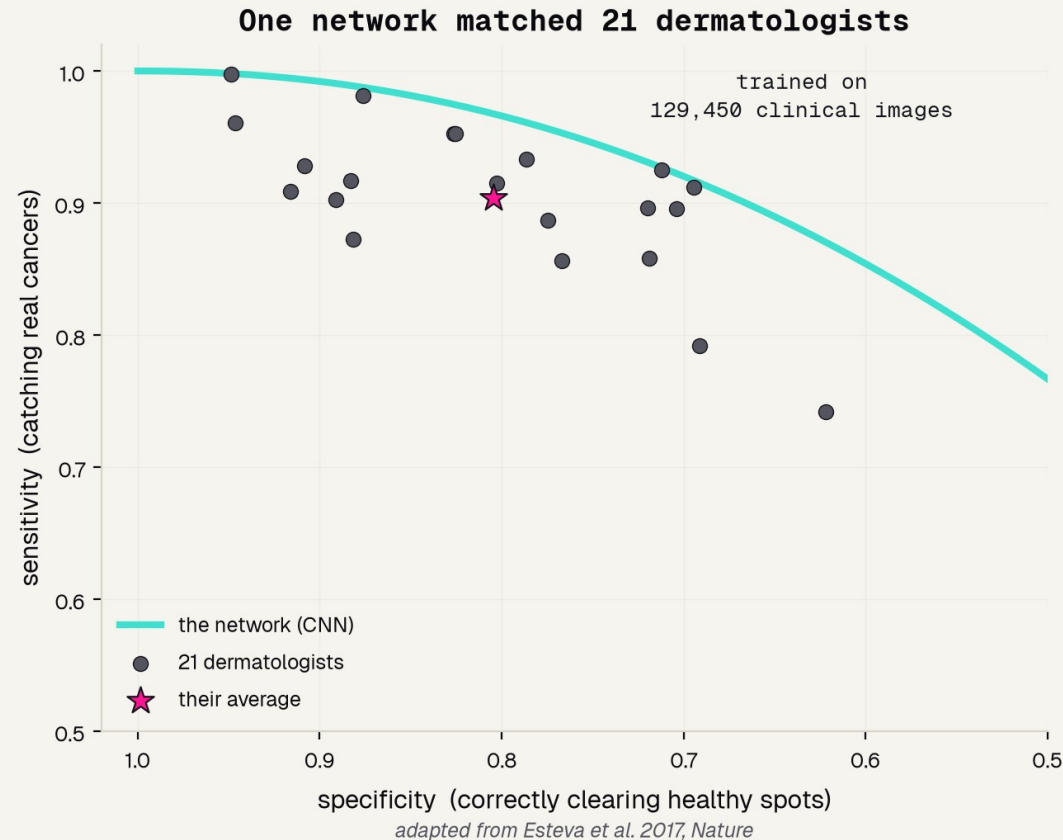
TWO MISTAKES, ONE IS WORSE

A false negative -- a real melanoma called harmless -- lets a cancer keep growing, and can be fatal.

A false positive is just one extra check-up. The costs are wildly lopsided.

So we do not chase balanced accuracy. We deliberately tilt the screen toward catching cancers.

Machines already match dermatologists



129,450

clinical images it learned from

21

board-certified dermatologists it matched

2017

the year this stopped being science fiction

THE DATA

Why HAM10000

10,015

real dermatoscopy images -- the kind a clinic actually captures, not cartoons

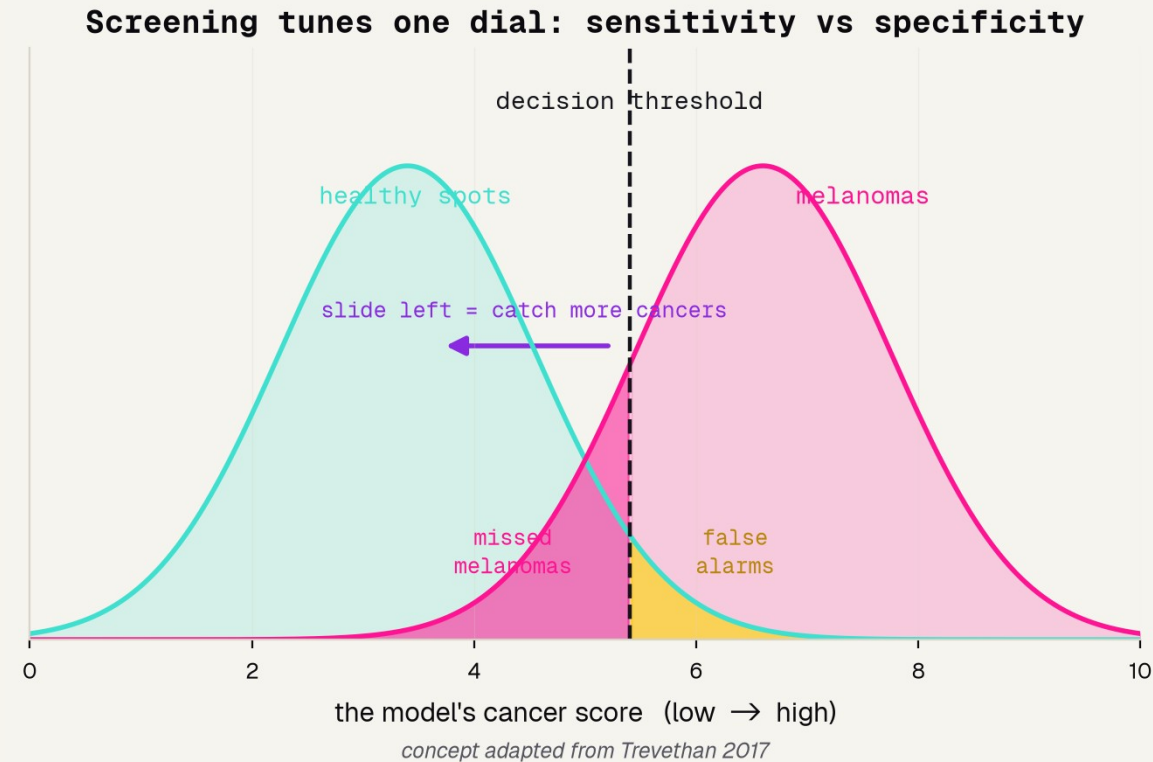
7

lesion types, which we collapse to one screening question: melanoma vs. everything else

11%

are melanoma -- the rare, deadly class -- and every image records age and sex, so we can audit fairness

Accuracy is a trap; sensitivity is the job



89% / 0%

a lazy "always not melanoma" model: 89% accurate, catches ZERO cancers

the fix: one dial

slide the decision threshold left to catch more melanomas -- we grade on sensitivity, not accuracy

THE MODEL

Borrow a brain: transfer learning

PRETRAINED

a network already trained on millions of everyday photos
-- cats, cars, faces

it already knows edges,
colors, shapes



FINE-TUNE

reuse those edge and color features, then train on ~10,000 HAM10000 skin images

it learns melanoma color
and border



SCREEN

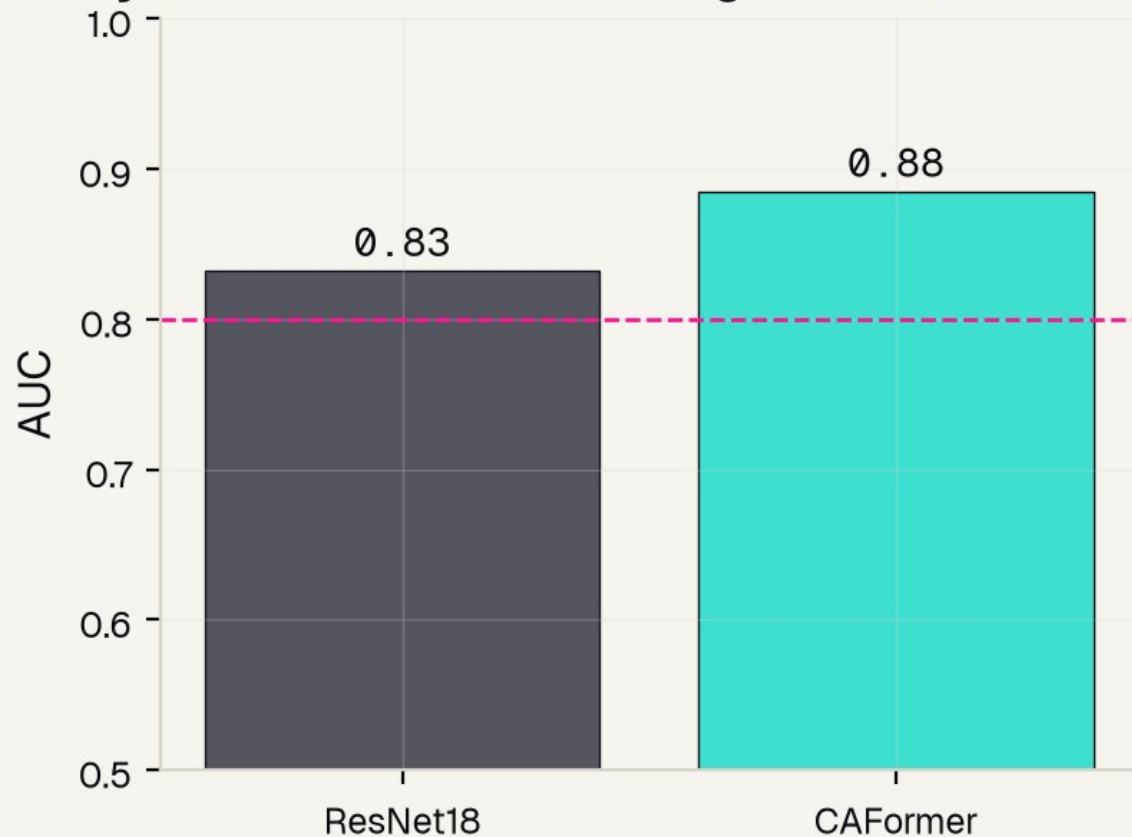
ask the fine-tuned network for one call on a new skin spot

one probability, 0 to 1

The reused knowledge is why a model can learn skin cancer from thousands, not millions, of photos.

Why CAFormer, not ResNet18

Why CAFormer: a stronger backbone wins



WE TESTED, WE DID NOT GUESS

0.88

CAFormer AUC -- vs 0.83 for a plain ResNet18.
Both clear 0.8, but the stronger backbone wins.

The winner is chosen on held-out data, never on a hunch.

Model and data processing

1

MODEL

caformer_s18, pretrained, backbone FROZEN, a fresh 2-class head trained on top

2

RESIZE

every image decoded to color and resized to 224x224 pixels

3

NORMALIZE

pixels rescaled to a fixed mean/scale, the range the backbone was trained on

4

CLASS-WEIGHT

the loss up-weights the rare melanomas, so the model cannot ignore them

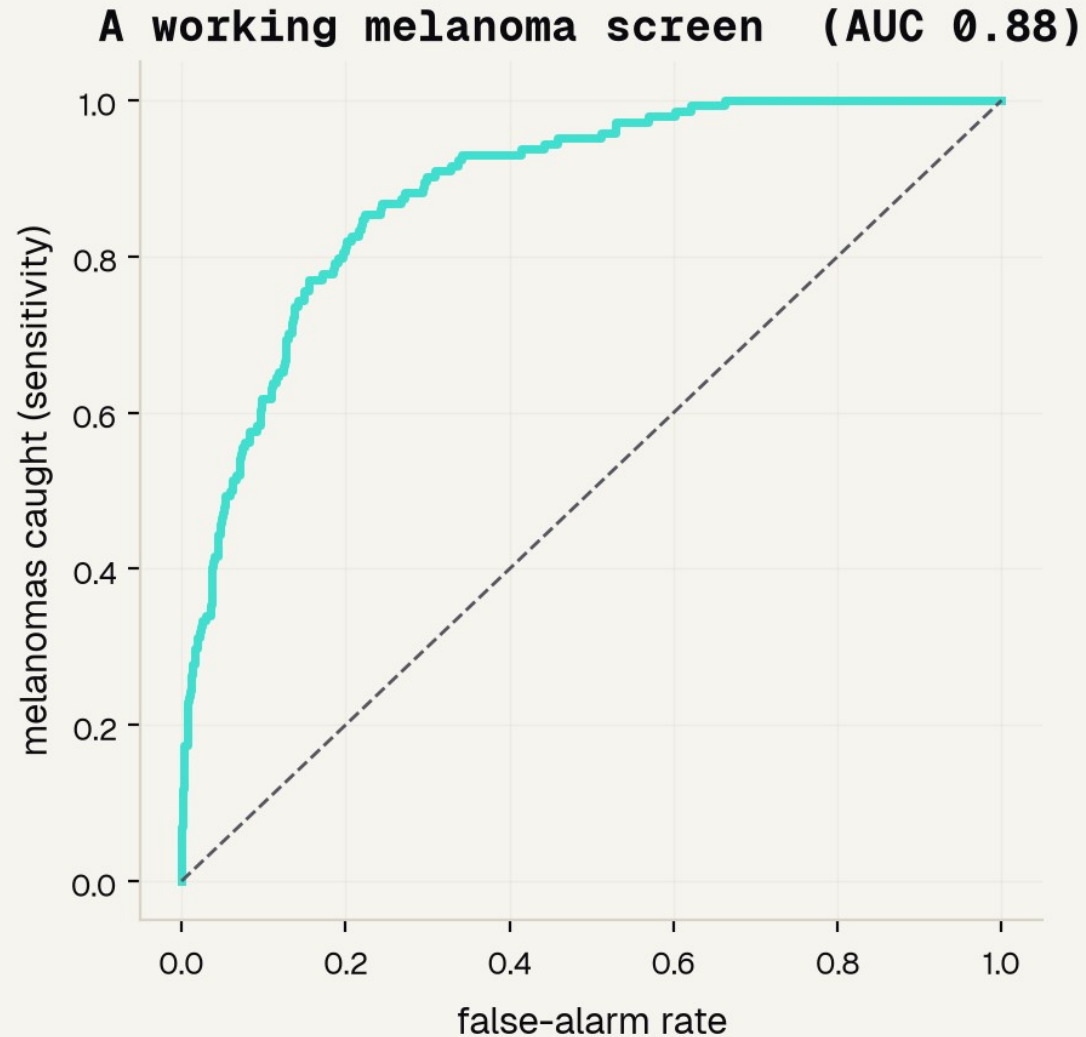
5

SPLIT

separate train / val / test; every number we report is on the held-out test set

RESULTS

A working screen



0.885

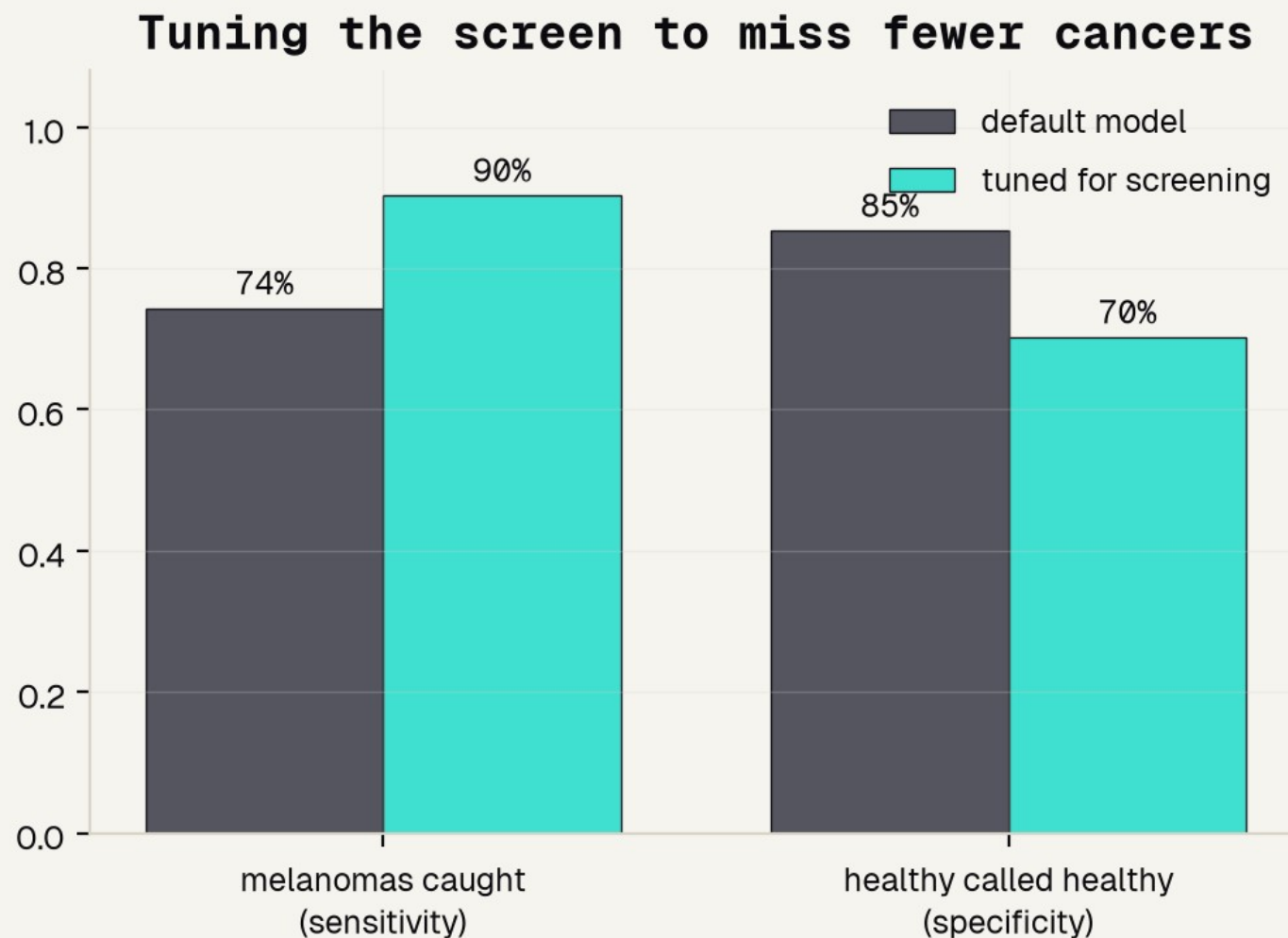
AUC -- the area under the ROC curve

The ROC curve sweeps every cut-off and plots melanomas caught against false alarms. AUC is the area underneath.

0.5 = a coin flip 1.0 = perfect

The curve hugs the top-left corner -- a genuinely working screen.

Tuning to miss fewer cancers



74% → 90%

melanomas caught (recall), after we lower the threshold

THE COST WE ACCEPT

Specificity slips 85% → 70%: more healthy spots get a second look. For a screen, that is the right trade -- a few extra check-ups to avoid a missed cancer.

At the screening setting

At the screening setting (catch 90%)

truth	not mel	801	340
	melanoma	14	130
		not mel	melanoma
		screen says	

130

melanomas CAUGHT -- the wins

14

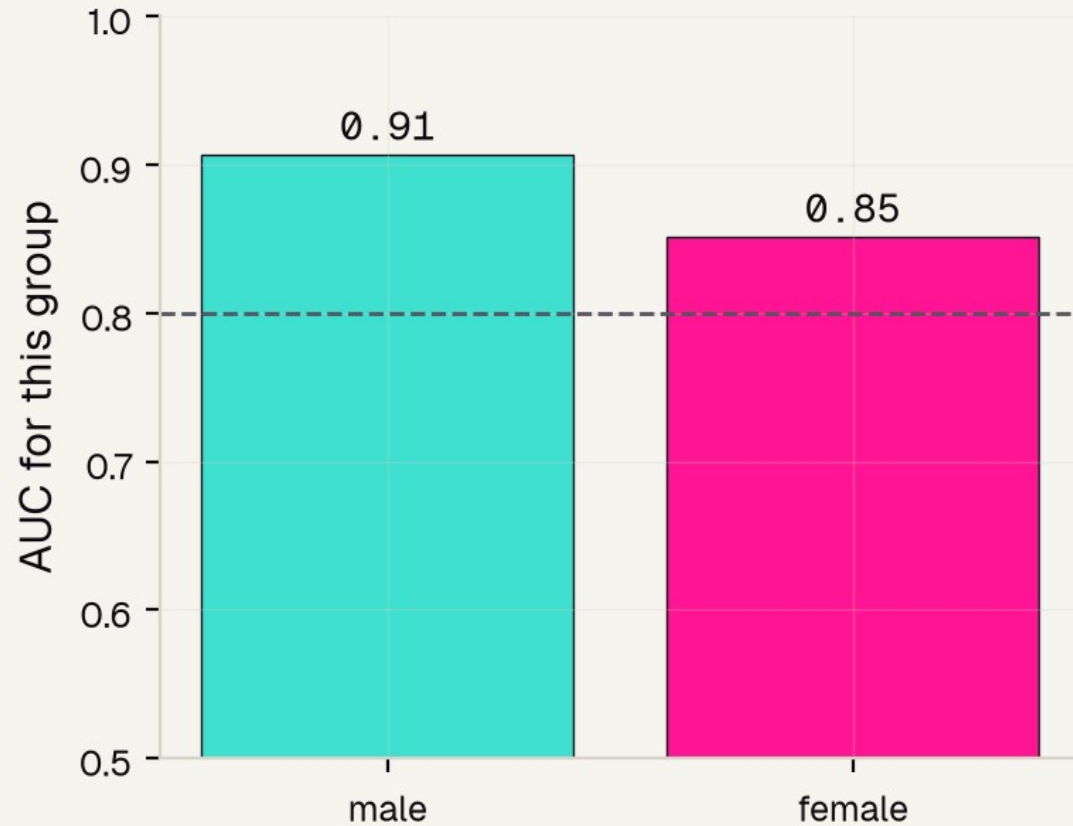
melanomas MISSED -- each one a real danger

340

false alarms -- each just an extra check-up

Does it work for women and men?

Fairness check: works for women and men



0.91

AUC for men

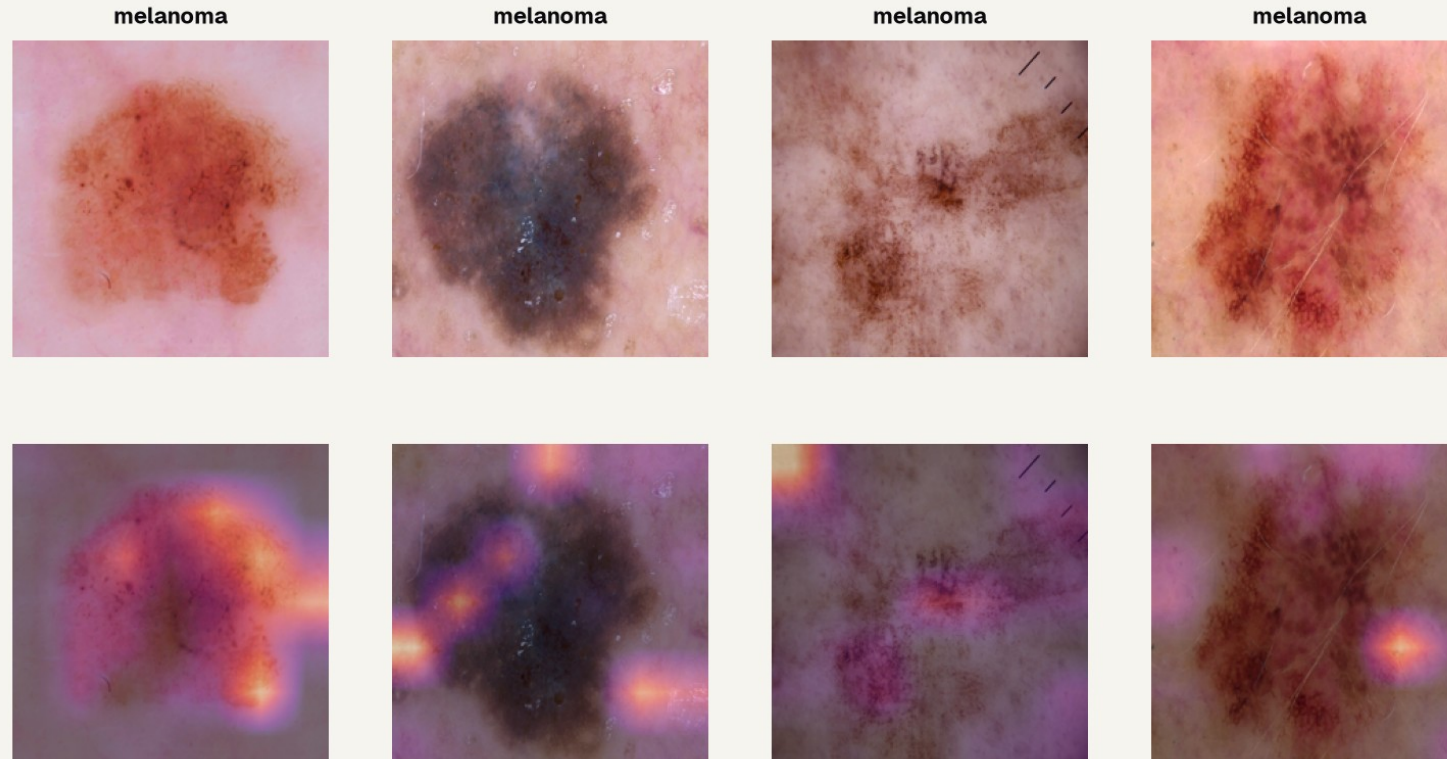
0.85

AUC for women

Mind the 0.06 gap. Both clear 0.8, but a screen you trust means naming where it is weaker -- and working to close it.

Where the screen looks

Where the screen looks (real melanomas)



Grad-CAM = feature importance for an image: the bright areas sit ON the pigmented lesion -- the same cues a dermatologist reads, not a hair, a ruler, or the frame corner.

BEING HONEST

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What it can and cannot do

IT WORKS

A real melanoma screen on real dermatoscopy: AUC 0.885, tuned to catch 90% of melanomas, and it looks at the lesion for the right reasons.

AUC 0.885 · recall 90%

BUT IT IS UNEVEN

A real gap by sex (0.85 vs 0.91), and HAM10000 skews toward lighter skin -- so performance on darker skin is unproven here.

a real 0.06 gap by sex

A REAL TOOL

Would train on more diverse patients, be validated across many hospitals, and keep a dermatologist in the loop before touching a patient.

not ready for a patient

References

LANDMARKS & DATA

- [1] Esteva et al. 2017, Nature -- one network matched 21 dermatologists.
- [2] Tschandl, Rosendahl & Kittler 2018, Sci Data -- the HAM10000 dataset.
- [3] Haenssle et al. 2018, Ann Oncol -- CNN vs 58 dermatologists on melanoma.
- [4] Tschandl et al. 2019, Lancet Oncol -- machines vs human readers, international.

METHODS & PRIMERS

- [5] Kim et al. 2022, BMC Med Imaging -- transfer learning for medical images.
- [6] Trevethan 2017, Front Public Health -- sensitivity, specificity, and their pitfalls.
- [7] Wei et al. 2024, Front Med -- AI and skin cancer: how well it works and the gaps.

The honest bottom line

A screen earns trust by catching the dangerous case.

Tune the threshold on purpose, check who it works for -- then use a validated tool and a dermatologist to actually screen a patient.

Thank You

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夢想